Lecture 15

Möbius maps

Objectives. *Fractional-linear maps describe all holomorphic automorphisms of the disk and the upper half-plane. We first describe their action on the sphere. The Schwarz lemma then reduces the disk classification to rotations.*

By the end of this lecture you will be able to:

- Compute a Möbius map at finite points and at infinity.
- Use the cross ratio to specify the images of three points.
- Prove the Schwarz lemma, including its equality cases.
- Classify the holomorphic automorphisms of the disk and half-plane.

15.1 Fractional-linear maps on the sphere

Recall that the Riemann sphere is $\widehat{\mathbb{C}} = \mathbb{C} \cup \{\infty\}$. Its coordinate near infinity is $w = 1/z$. A *Möbius map* is the map of this sphere given at finite points by

$$T_g(z) = \frac{\alpha z + \beta}{\gamma z + \delta}, \qquad g = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}, \qquad \alpha\delta - \beta\gamma \neq 0.$$

Here α, β, γ and δ denote matrix entries. In this formula γ is not a contour and δ is not a distribution. If $\gamma \neq 0$, define $T_g(-\delta/\gamma) = \infty$ and $T_g(\infty) = \alpha/\gamma$. If $\gamma = 0$, define $T_g(\infty) = \infty$. The determinant condition prevents numerator and denominator from vanishing together. Near the pole use $1/T_g$ as the target coordinate. Near infinity replace z by $1/w$. These expressions are holomorphic in their respective charts. Thus the formulas define a holomorphic map of the whole sphere.

Multiplying all entries of g by a nonzero scalar does not change T_g. Choose a square root of its determinant and divide by that root. The new matrix has determinant one. The two root choices give opposite matrices. Write $\mathrm{SL}(2,\mathbb{C})$ for the determinant-one matrices, and write $\mathrm{PSL}(2,\mathbb{C}) = \mathrm{SL}(2,\mathbb{C})/\{I, -I\}$. An *action* assigns a bijection T_g to each group element so that T_I is the identity and $T_g \circ T_h = T_{gh}$. It is *faithful* if only the identity group element acts as the identity map.

Theorem 15.1. *The rule $g \cdot z = T_g(z)$ defines a faithful action of $\mathrm{PSL}(2,\mathbb{C})$ on $\widehat{\mathbb{C}}$ by holomorphic bijections.*

Proof. Substitution of the fractional formulas gives $T_g \circ T_h = T_{gh}$. Indeed, the numerator entries of the composite are the first row of gh, and its denominator entries are the second row. Equality at exceptional points follows by continuity in the sphere charts. The inverse is $T_{g^{-1}}$. Hence every map is a holomorphic bijection with holomorphic inverse.

If $T_g(z) = z$ for all finite points away from the possible pole, then

$$\gamma z^2 + (\delta - \alpha)z - \beta = 0.$$

A polynomial vanishing at infinitely many points has every coefficient zero. Thus $\gamma = \beta = 0$ and $\delta = \alpha$. The determinant-one condition gives $\alpha = 1$ or $\alpha = -1$. Exactly I and $-I$ act trivially, which proves faithfulness of the quotient. ∎

For example, the matrix $\left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$ gives $T(z) = -1/z$. This map interchanges zero and infinity. Its square is the identity map, although the square of its matrix is $-I$. This example explains why the quotient by $\{I, -I\}$ occurs.

15.2 The cross ratio

For distinct finite points define their *cross ratio* by

$$[z_1, z_2; z_3, z_4] = \frac{(z_1 - z_3)(z_2 - z_4)}{(z_1 - z_4)(z_2 - z_3)}.$$

When one point is infinity, take the corresponding limit. The order of the four points matters. For instance, $[0, 1; 2, 3] = (-2)(-2)/((-3)(-1)) = 4/3$.

Proposition 15.2. *Every Möbius map preserves the cross ratio, using limits when a point or its image is infinity. Given two ordered triples of distinct sphere points, there is exactly one Möbius map taking the first triple to the second.*

Proof. At ordinary finite points subtraction gives

$$T_g(z) - T_g(v) = \frac{(\alpha\delta - \beta\gamma)(z - v)}{(\gamma z + \delta)(\gamma v + \delta)}.$$

Insert this expression in the cross ratio. Each determinant and each linear denominator cancels. Limits prove the exceptional cases.

For a triple (a, b, c), the map

$$Q_{a,b,c}(z) = \frac{(z - b)(c - a)}{(z - a)(c - b)}$$

takes (a, b, c) to $(\infty, 0, 1)$. Limits give the same construction if one point is infinity. Its numerator and denominator have distinct zeros, so its determinant is nonzero. Composing two such maps, with one inverted, gives existence for arbitrary triples. A Möbius map fixing infinity has $\gamma = 0$. If it also fixes zero and one, its formula is $z \mapsto z$. Conjugating by the constructed maps gives uniqueness. ∎

The triple construction has a useful real version. If all three points in each triple lie in $\mathbb{R} \cup \{\infty\}$, the displayed formulas have real coefficients. Thus the interpolating map has a real matrix, up to an overall scalar. This consequence will identify the half-plane automorphisms.

To construct a map between prescribed triples, first send the source triple to the standard triple used in the proof. Send the target triple to that same standard triple by a second map. Apply the first map. Then apply the inverse of the second. Substituting each prescribed point checks the order of composition, including any value at infinity.

15.3 The Schwarz lemma

Write $\mathbb{D} = \{z \in \mathbb{C} : |z| < 1\}$ for the unit disk. A bijective holomorphic map with holomorphic inverse is called *biholomorphic*. An *automorphism* of a domain is a biholomorphic map from the domain to itself. We first control a holomorphic self-map with a fixed point at zero.

Theorem 15.3 (Schwarz). *Let $f : \mathbb{D} \to \mathbb{D}$ be holomorphic and satisfy $f(0) = 0$. Then $|f(z)| \le |z|$ for $z \in \mathbb{D}$ and $|f'(0)| \le 1$. If equality holds in the first inequality at some $z \neq 0$, or if $|f'(0)| = 1$, then $f(z) = e^{i\theta} z$ for a real angle θ. Conversely, each such rotation has both equalities.*

Proof. By the Taylor theorem, Theorem 10.3, the quotient $f(z)/z$ has a holomorphic extension h to zero, with $h(0) = f'(0)$. For $0 < r < 1$, its boundary values satisfy $|h(z)| \le 1/r$ on $|z| = r$. The maximum-modulus principle, Theorem 10.7, gives the same bound inside that circle. Fix $z \in \mathbb{D}$ and let $r \uparrow 1$ with $r > |z|$. This proves $|h(z)| \le 1$, including at zero.

Either equality hypothesis says that h attains modulus one at an interior point. The equality clause of the maximum-modulus principle makes h constant. Its constant value has modulus one and is therefore $e^{i\theta}$ for some $\theta \in \mathbb{R}$. Substitution verifies the converse. ∎

For example, $f(z) = z^2$ maps the disk to itself and fixes zero. The inequality is strict at every nonzero point because $0 < |z| < 1$ implies $|z|^2 < |z|$. This map cannot be an automorphism: it identifies z and $-z$.

15.4 Automorphisms of the disk

For a point $a \in \mathbb{D}$, define the disk factor b_a by

$$b_a(z) = \frac{z - a}{1 - \overline{a} z}.$$

Its denominator is nonzero on the closed unit disk, since $|a| < 1$. Direct expansion gives

$$1 - |b_a(z)|^2 = \frac{(1 - |a|^2)(1 - |z|^2)}{|1 - \overline{a} z|^2}. \tag{15.1}$$

Thus b_a maps $\mathbb{D}$ into itself and takes a to zero. Solving $\zeta = b_a(z)$ gives

$$b_a^{-1}(\zeta) = \frac{\zeta + a}{1 + \overline{a}\zeta} = b_{-a}(\zeta).$$

This inverse also maps the disk into itself by (15.1). Both formulas are holomorphic there.

Theorem 15.4. *The automorphisms of $\mathbb{D}$ are exactly*

$$f(z) = e^{i\theta} \frac{z - a}{1 - \overline{a} z}, \qquad a \in \mathbb{D}, \quad \theta \in \mathbb{R}.$$

Proof. Let $a = f^{-1}(0)$ and set $h = f \circ b_a^{-1}$. This automorphism fixes zero, and its inverse also fixes zero. Applying Schwarz to h and to h^{-1} gives

$$|h(z)| \le |z| = |h^{-1}(h(z))| \le |h(z)|.$$

Equality at a nonzero point makes h a rotation. Hence $f = e^{i\theta} b_a$. Conversely, rotations and the already verified maps b_a are automorphisms, so every displayed map is one. ∎

For example, taking $a = 1/2$ and $\theta = 0$ gives $f(z) = (2z - 1)/(2 - z)$. This map takes $1/2$ to zero and zero to $-1/2$. It extends to the unit circle and fixes its real endpoints 1 and -1. The interior points change while those boundary points stay fixed.

15.5 Automorphisms of the upper half-plane

Write $\mathbb{H} = \{z \in \mathbb{C} : \operatorname{Im} z > 0\}$. The Cayley map $C : \mathbb{H} \to \mathbb{D}$ is given by

$$C(z) = \frac{z - \mathrm{i}}{z + \mathrm{i}}, \qquad C^{-1}(\zeta) = \mathrm{i}\frac{1 + \zeta}{1 - \zeta}.$$

The inequality $|z - \mathrm{i}| < |z + \mathrm{i}|$ is equivalent to $\operatorname{Im} z > 0$. The sphere bijection in Theorem 15.1 therefore maps the half-plane onto the disk. Its displayed inverse is holomorphic there, so this restriction is biholomorphic. It takes i to zero and $2\mathrm{i}$ to $1/3$. The following diagram records the boundary points as sphere points.

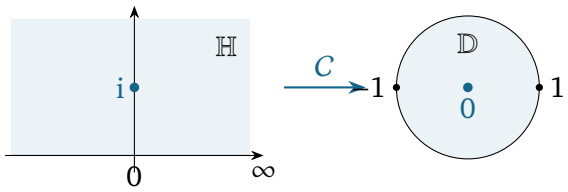


Figure 15.1: The Cayley map takes the half-plane to the disk. On the boundary sphere it takes 0 to -1 and infinity to 1.

Theorem 15.5. *The automorphisms of $\mathbb{H}$ are exactly*

$$T(z) = \frac{\alpha z + \beta}{\gamma z + \delta}, \qquad \alpha, \beta, \gamma, \delta \in \mathbb{R}, \quad \alpha\delta - \beta\gamma = 1.$$

The two matrices differing by a sign give the same map.

Proof. For real coefficients with determinant one, subtraction from the complex conjugate gives

$$\operatorname{Im} T(z) = \frac{\operatorname{Im} z}{|\gamma z + \delta|^2}.$$

The denominator has no zero in $\mathbb{H}$. The inverse matrix has real entries and determinant one, so this formula also proves surjectivity onto $\mathbb{H}$.

Conversely, conjugate an automorphism of $\mathbb{H}$ by C. The disk classification shows that the resulting map is Möbius and extends bijectively to the unit circle. The Cayley map sends the extended real axis onto that circle: equality of the two distances holds exactly on the real axis, and infinity maps to one. Conjugating back gives a Möbius map taking $\mathbb{R} \cup \{\infty\}$ bijectively to itself. Choose three distinct points on that boundary. Their images are also real sphere points. Real three-point interpolation supplies a real matrix giving the same images. Uniqueness of interpolation makes its map the given map. The imaginary-part formula with determinant d has numerator $d \operatorname{Im} z$. Preservation of $\mathbb{H}$ forces $d > 0$. Dividing the real matrix by $\sqrt{d}$ normalises its determinant to one. The kernel computation in Theorem 15.1 leaves precisely the sign ambiguity. ∎

For example, $T(z) = z + s$ with $s \in \mathbb{R}$ translates the half-plane. Its matrix is $\left(\begin{smallmatrix} 1 & s \\ 0 & 1 \end{smallmatrix}\right)$. The map $T(z) = -1/z$ is another example, since its imaginary part is $\operatorname{Im} z/|z|^2 > 0$ there.

Summary.

- Determinant-one matrices act on the sphere, with kernel $\{I, -I\}$. Möbius maps preserve the stated ordered cross ratio.
- Schwarz bounds a disk self-map fixing zero by $|f(z)| \le |z|$. Equality away from zero, or $|f'(0)| = 1$, forces a rotation.
- Every disk automorphism is $e^{i\theta} b_a$. Every half-plane automorphism has a real determinant-one matrix.

The Cayley map transfers the two classifications to one another.

References

E. M. Stein and R. Shakarchi, *Complex Analysis*, 2003, Chapter 8, §2, pp. 218–223. Their disk factor and Cayley map are the negatives of ours. M. J. Ablowitz and A. S. Fokas, *Introduction to Complex Variables and Applications*, 2021, §5.7, pp. 350–358. Their cross-ratio ordering differs from the convention used here.